Sean Chen

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Education

University of California, Berkeley Berkeley, CA
B.S. in Mechanical Engineering *Expected May 2028*
Coursework: Dynamics, Mechanical Behavior, Statistics & Data Science, Statics, Mechanics of Materials, MATLAB

Clubs/Research: CalSol, Gamlean Project (Prof. George Anwar), VP and Founder of SASE @ Mt. SAC, Chinese Student Association (CSA)

Experience

Mechanical Design Member – Dynamics Subteam *August 2026 – Present*
CalSol Solar Vehicle Team *UC Berkeley*
- Evaluated multiple wheel spoke geometries in SolidWorks and selected an eight-spoke 6061-T6 aluminum design to improve load distribution while minimizing mass
- Derived a 1,558 N bump load and 169 Nm braking torque from vehicle weight and 2g vertical/1g longitudinal acceleration conditions; validated structural performance using static FEA, mesh convergence, and hand calculations
- Reduced modeled wheel spoke mass by 16.8% (698 g to 581 g) compared to the previous year's design while achieving FEA-predicted factors of safety of 1.6 under bump loading and 3.9 under braking

Project Lead *June 2026 – August 2026*
MathWorks Modeling Challenge *Remote*
- Built a quarter-car suspension model using MATLAB, Simulink, and Simscape; automated 27 simulations across 9 spring-damper configurations and 3 road conditions
- Compared ride comfort, suspension travel, and tire deflection to select the best overall suspension setup, improving the normalized performance score by 2.9%
- Optimized suspension with a 25% increase in sprung mass, meeting performance requirements across all 3 simulated road conditions

Mechanical Design Lead – Student Research Project *February 2026 – June 2026*
Mt. San Antonio College *Walnut, CA*
- Led mechanical design within a 5-person team developing a tethered weather-balloon system to collect vertical air-quality measurements at altitudes up to 300 meters
- Performed FEA and shape optimization in Fusion 360, reducing payload mass by 25.9% (353 g to 261 g) to minimize helium cost
- Developed 3D CAD models, 2D engineering drawings, GD&T, and a Bill of Materials with dimensions, tolerances, material specifications, and assembly requirements
- Validated the payload through 20 N tether-load testing with no measured deformation; verified electronics fit and sensor functionality

Projects

Autonomous Object-Tracking System | *Fusion 360, Raspberry Pi, Python* *October 2025 – February 2026*
- Designed a 2-axis camera gimbal in Fusion 360, calculating torque requirements using camera mass and gear geometry to select a stepper motor and develop the gear-driven movement system
- Conducted range-of-motion, repeatability, and structural stiffness tests to evaluate gimbal performance and identify improvements to mechanical alignment
- Integrated YOLOv8 object-detection outputs with motor commands to convert detected target position into automated gimbal movement

Folding Knife | *SolidWorks, 3D Printing, Fabrication* *April 2026 – May 2026*
- Iterated through three SolidWorks knife designs and replaced an unsuccessful friction design with a spring-steel back-lock mechanism
- Fabricated and finished components using an angle grinder, belt sander, Dremel, and created a bevel-grinding jig to maintain a 13-degree blade angle
- Developed and executed a heat-treatment process for the annealed A2 tool-steel blade, achieving a final hardness of 58 HRC verified through Rockwell C testing

Technical Skills

CAD & Engineering: SolidWorks, Fusion 360, Siemens NX, Onshape, Engineering Drawings, GD&T
Simulation and Programming: FEA, Ansys, MATLAB, Simulink, Simscape, Python, Arduino, Raspberry Pi
Fabrication and Testing: 3D Printing, Angle Grinder, Belt Sander, Dremel, Heat Treatment, Rockwell C Hardness Testing
Interests & Hobbies: Hiking, Photography, Concerts, Golfing, Video Games